A SURJECTIVE ETALE MAP IS COVERED BY A FINITE FLAT ONE FACTORING THROUGH IT LOCALLY

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Local standard etaleness over a finite cover. Used at the start of the counted proof (lines 40367-40370). Produces finitely many elements generating the unit ideal over which the map is standard etale.
- Products of finite flat algebras. Used to assemble the global statement (lines 40370-40390). Combines the local solutions into one finite flat algebra mapping surjectively to the base.

Source reference. Original lemma 00E8: sources/stacks-algebra/pinned-original-source.tex, source lines 3469–3469. Navigation pointer for \ref{lemma-quasi-compact} used inside the counted span (line 40368); the argument that consumes it is retained in the span.

Source reference. Original lemma 00UF: sources/stacks-algebra/pinned-original-source.tex, source lines 40287–40287. Navigation pointer for \ref{lemma-standard-etale-finite-flat-Zariski} used inside the counted span (line 40378); the argument that consumes it is retained in the span.

Date: stacks-project:algebra.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 00UE: sources/stacks-algebra/pinned-original-source.tex, source lines 40061–40061. Navigation pointer for \ref{proposition-etale-locally-standard} used inside the counted span (line 40367); the argument that consumes it is retained in the span.

Lemma 0.1. *Let $R \to S$ be a ring map. Assume that*

(1) *$R \to S$ is étale, and*
(2) *$\mathrm{Spec}(S) \to \mathrm{Spec}(R)$ is surjective.*

Then there exists a ring map $R \to S'$ such that

(1) *$R \to S'$ is finite, finitely presented, and flat (in other words it is finite projective as an R-module),*
(2) *$\mathrm{Spec}(S') \to \mathrm{Spec}(R)$ is surjective,*
(3) *for every prime $\mathfrak{q}' \subset S'$ there exists a $g' \in S'$, $g' \notin \mathfrak{q}'$ such that the ring map $R \to S'_{g'}$ factors as $R \to S \to S'_{g'}$.*

Proof. By Proposition 00UE and the quasi-compactness of $\mathrm{Spec}(S)$ (see Lemma 00E8) we can find $g_1, \ldots, g_n \in S$ generating the unit ideal of S such that each $R \to S_{g_i}$ is standard étale. If we prove the lemma for the ring map $R \to \prod_{i=1,\ldots,n} S_{g_i}$ then the lemma follows for the ring map $R \to S$. Hence we may assume that $S = \prod_{i=1,\ldots,n} S_i$ is a finite product of standard étale morphisms.

For each i choose a ring map $R \to S'_i$ as in Lemma 00UF adapted to the standard étale morphism $R \to S_i$. Set $S' = S'_1 \otimes_R \ldots \otimes_R S'_n$; we will use the R-algebra maps $S'_i \to S'$ without further mention below. We claim this works. Properties (1) and (2) are immediate. For property (3) suppose that $\mathfrak{q}' \subset S'$ is a prime. Denote $\mathfrak{p}$ its image in $\mathrm{Spec}(R)$. Choose $i \in \{1, \ldots, n\}$ such that $\mathfrak{p}$ is in the image of $\mathrm{Spec}(S_i) \to \mathrm{Spec}(R)$; this is possible by assumption. Set $\mathfrak{q}'_i \subset S'_i$ the image of $\mathfrak{q}'$ in the spectrum of S'_i. By construction of S'_i there exists a $g'_i \in S'_i$ such that $R \to (S'_i)_{g'_i}$ factors as $R \to S_i \to (S'_i)_{g'_i}$. Hence also $R \to S'_{g'_i}$ factors as

$$R \to S_i \to (S'_i)_{g'_i} \to S'_{g'_i}$$

as desired. □